

E9 205 Machine Learning for Signal Processing

Dimensionality Reduction - I

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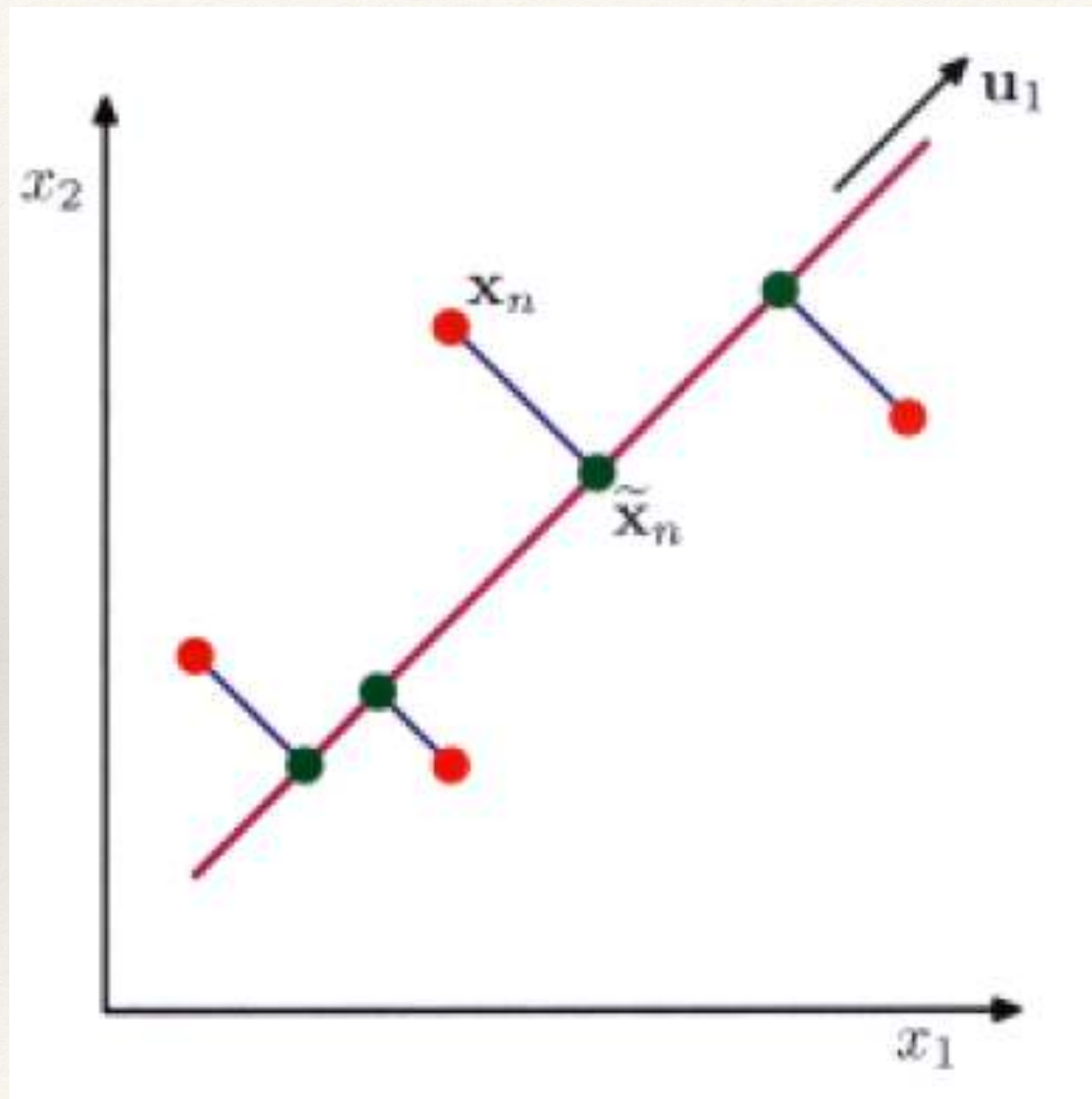
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Principal Component Analysis

- ❖ Reducing the data $\mathbf{x}_n$ of dimension D to lower dimension $M < D$
- ❖ Projecting the data into subspace which preserves maximum data variance
 - ❖ Maximize variance in projected space
- ❖ Equivalent formulated as minimizing the error between the original and projected data points.

Minimum Error Formulation - PCA



Principal Component Analysis

- ❖ First M eigenvectors of data covariance matrix

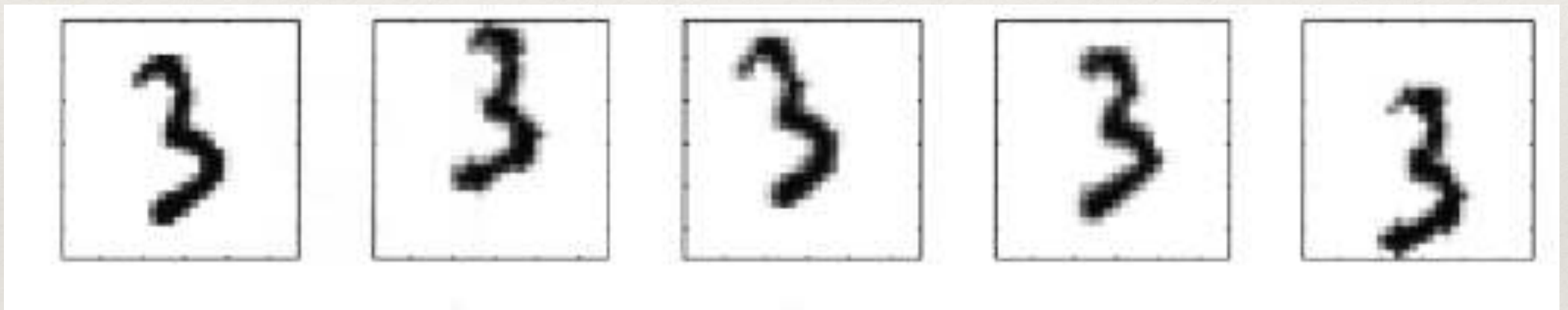
$$S = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$$

- ❖ Residual error from PCA

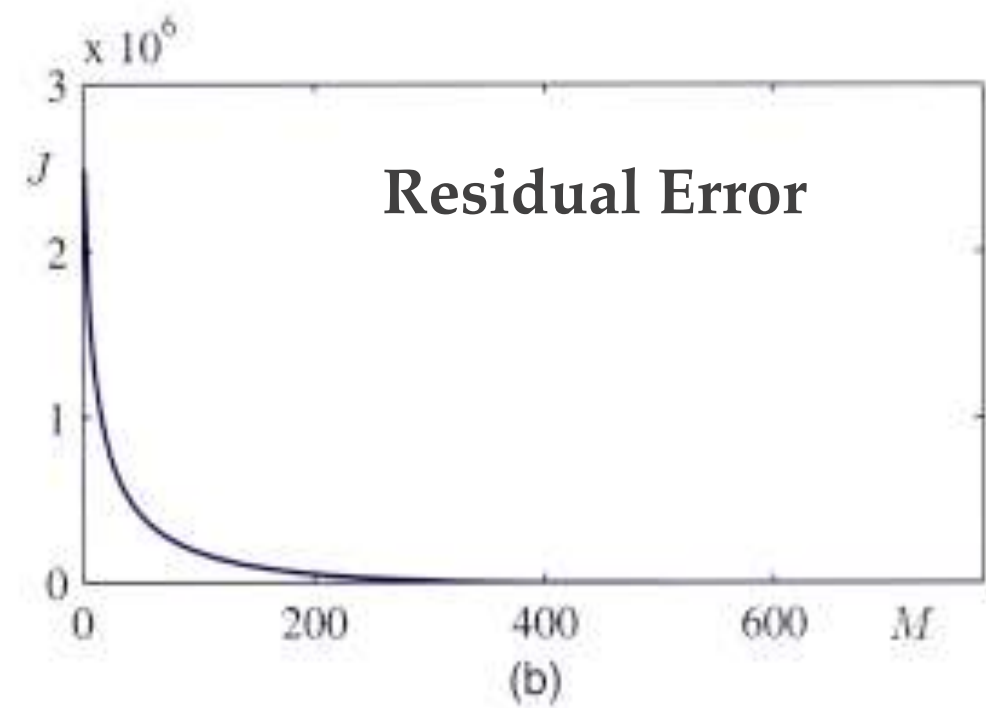
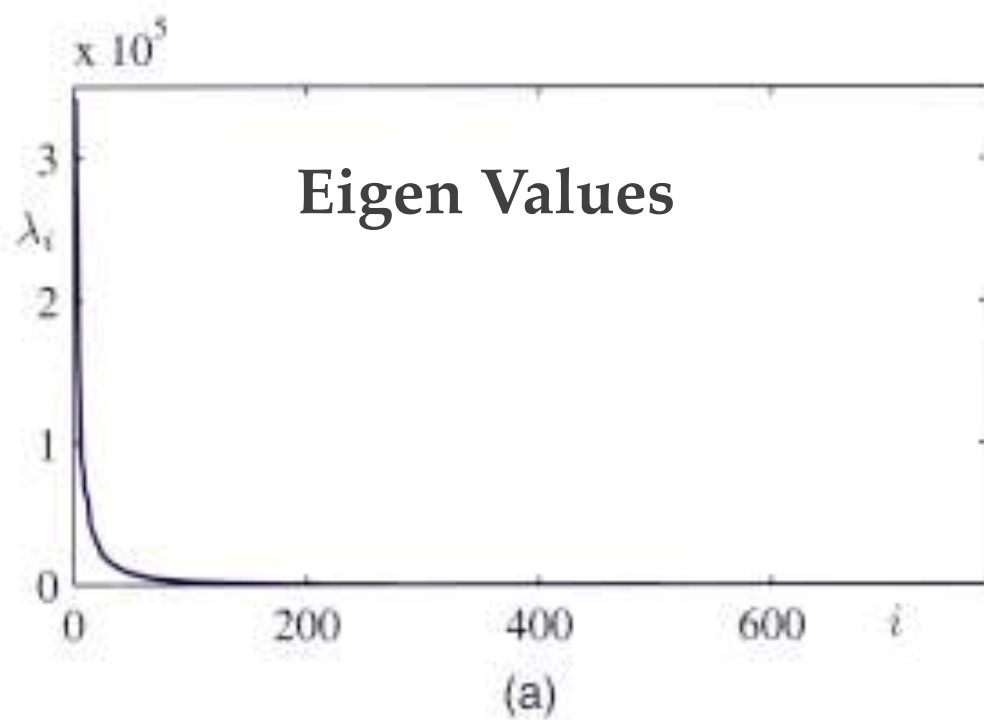
$$J = \sum_{i=M+1}^D \lambda_i$$

PCA

Handwritten digits used for PCA training...

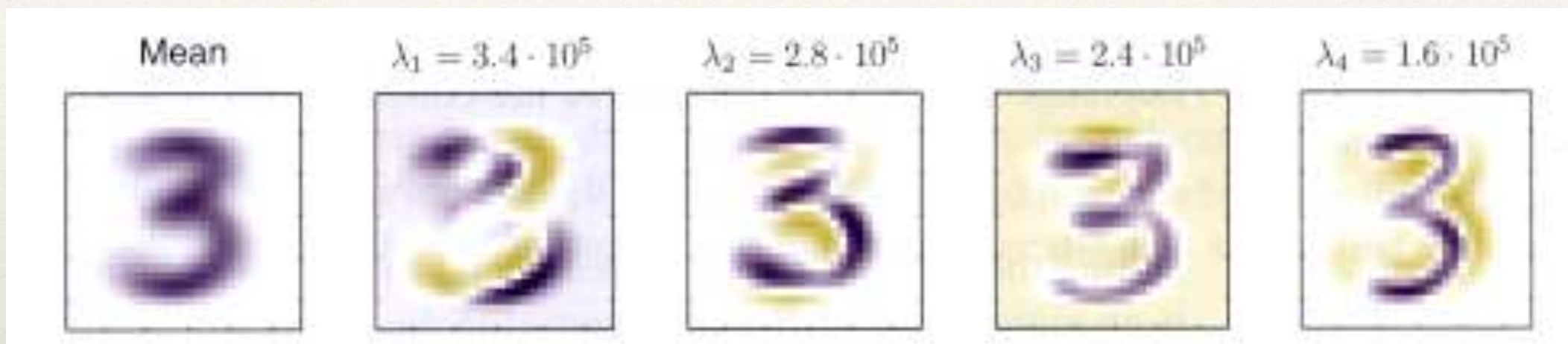


PCA

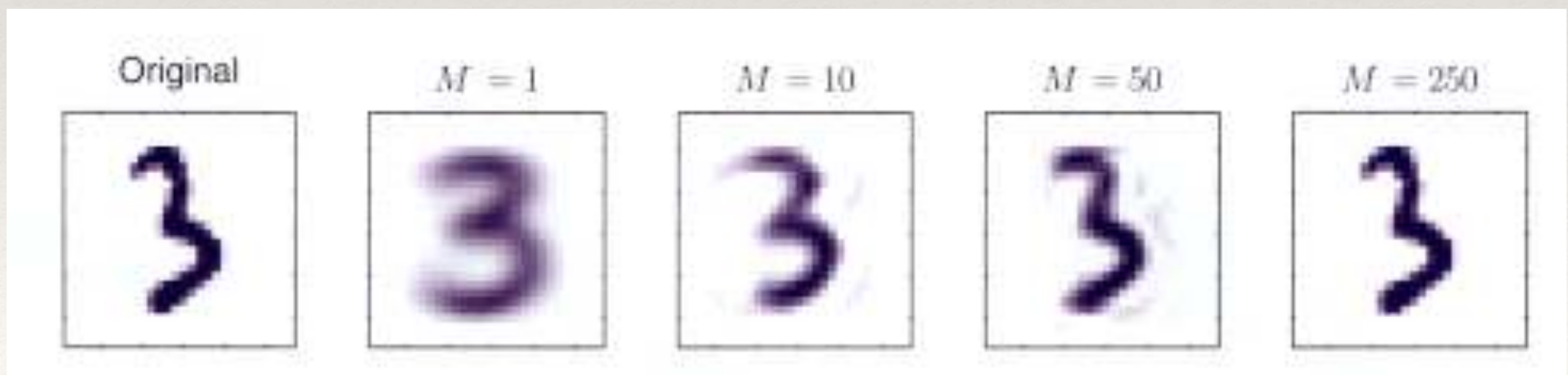


PCA - Reconstruction

Eigenvectors

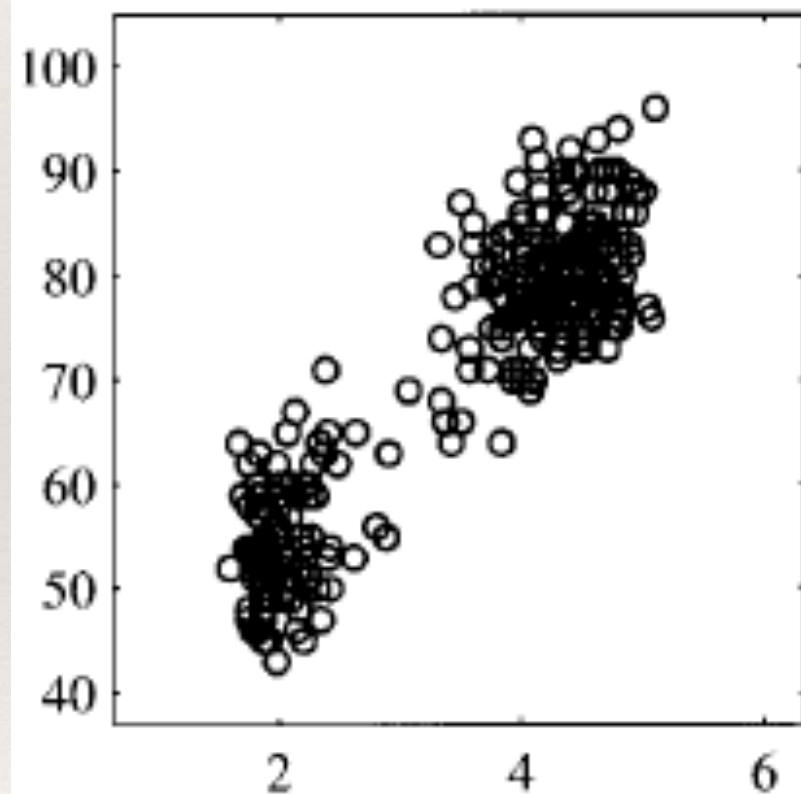


PCA - Reconstruction

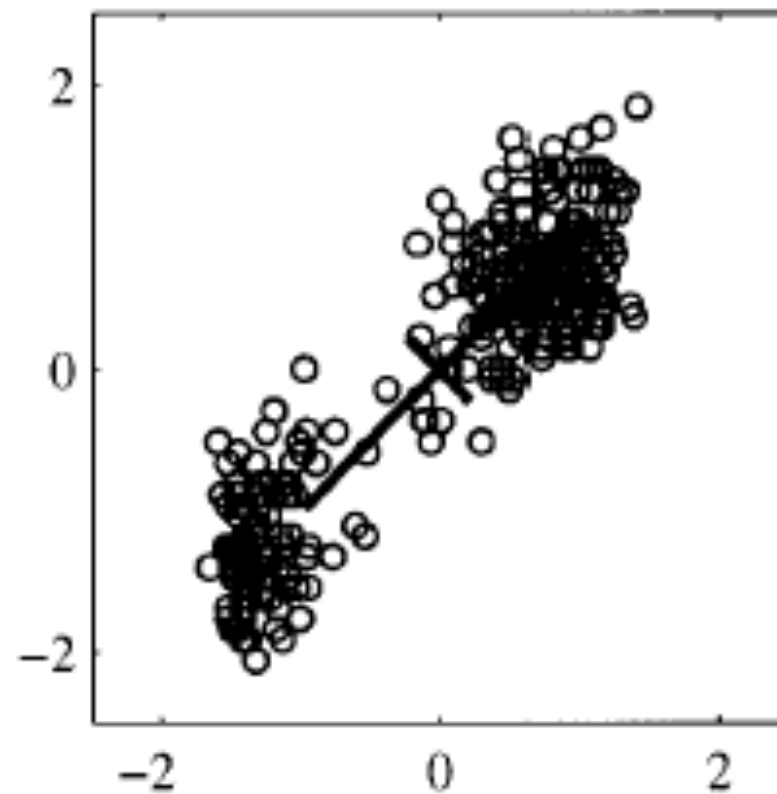


Whitening the Data

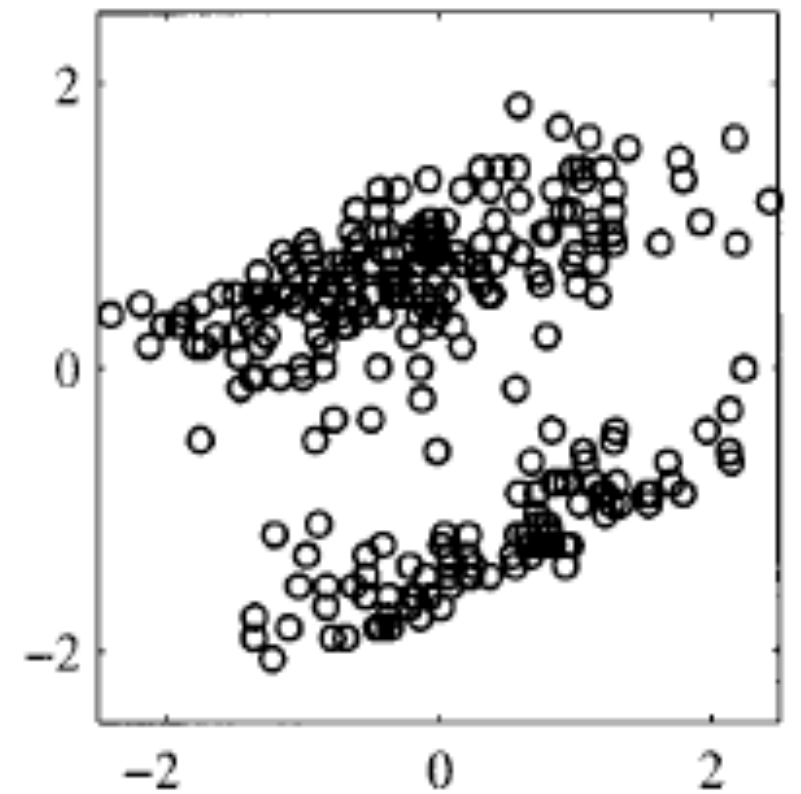
Original Data



Standardization

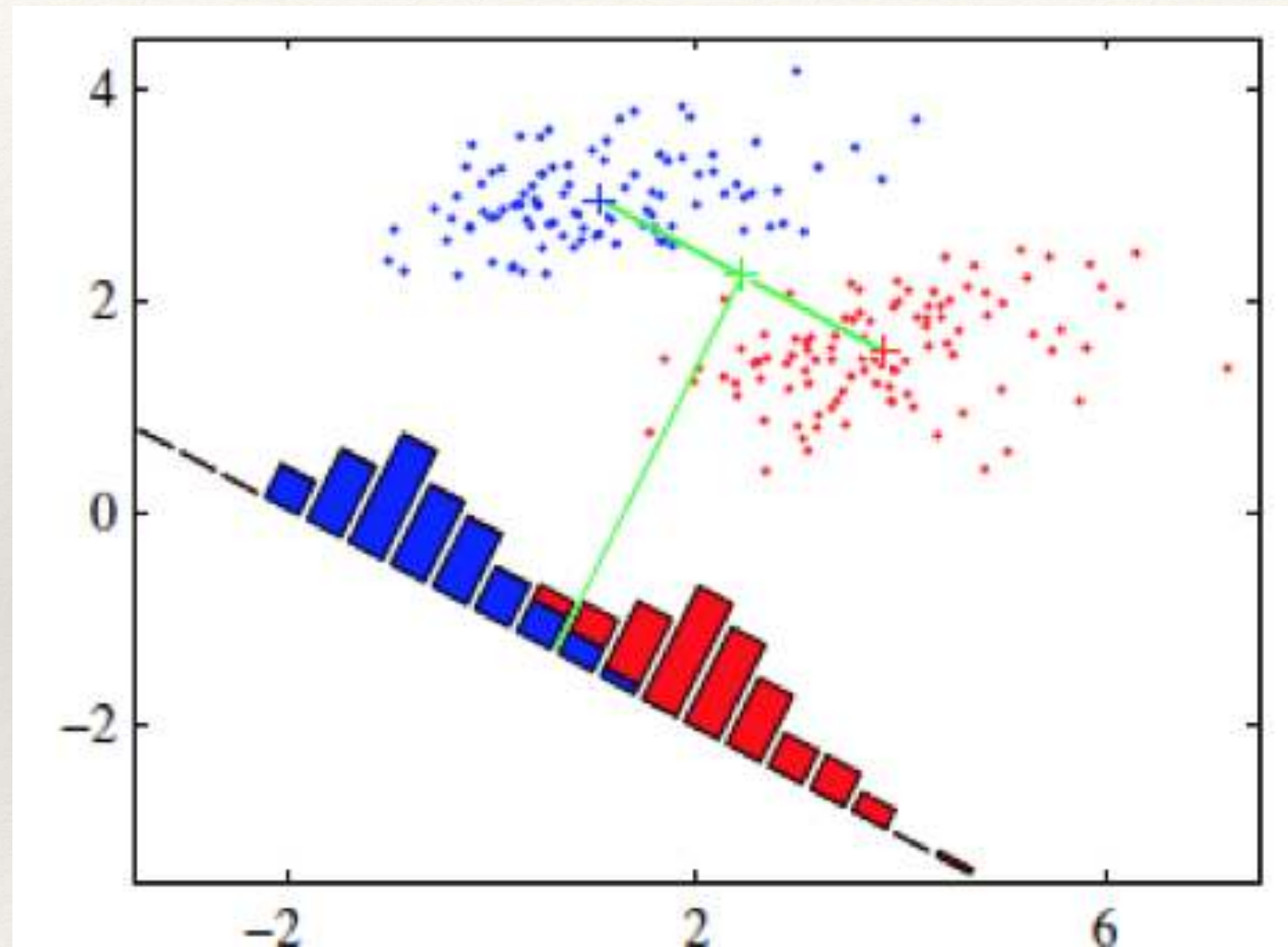


Whitening



Linear Discriminant Analysis

Without the Within Class Factor



Linear Discriminant Analysis

Find a linear transform $f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}$ with a criterion which maximizes the class separation

- Maximize the between class distance in the projected space while minimizing the within class covariance

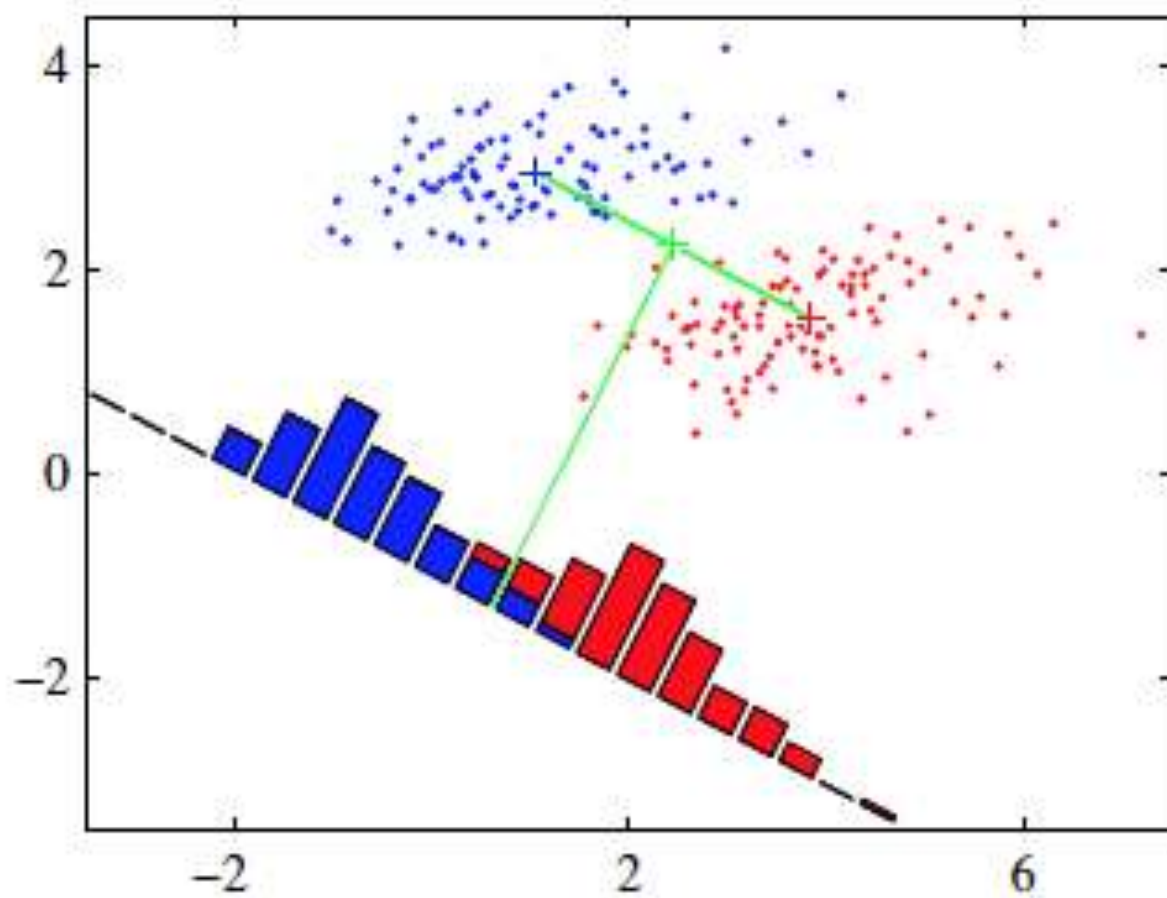
$$J = \frac{\mathbf{w}^T \mathbf{S}_b \mathbf{w}}{\mathbf{w}^T \mathbf{S}_w \mathbf{w}}$$

$$\mathbf{S}_b = \sum_{k=1}^K N_k (\mathbf{m}_k - \mathbf{m})(\mathbf{m}_k - \mathbf{m})^T \quad \mathbf{S}_w = \sum_{k=1}^K \sum_{n \in C_k} (\mathbf{x}_n - \mathbf{m}_k)(\mathbf{x}_n - \mathbf{m}_k)^T$$

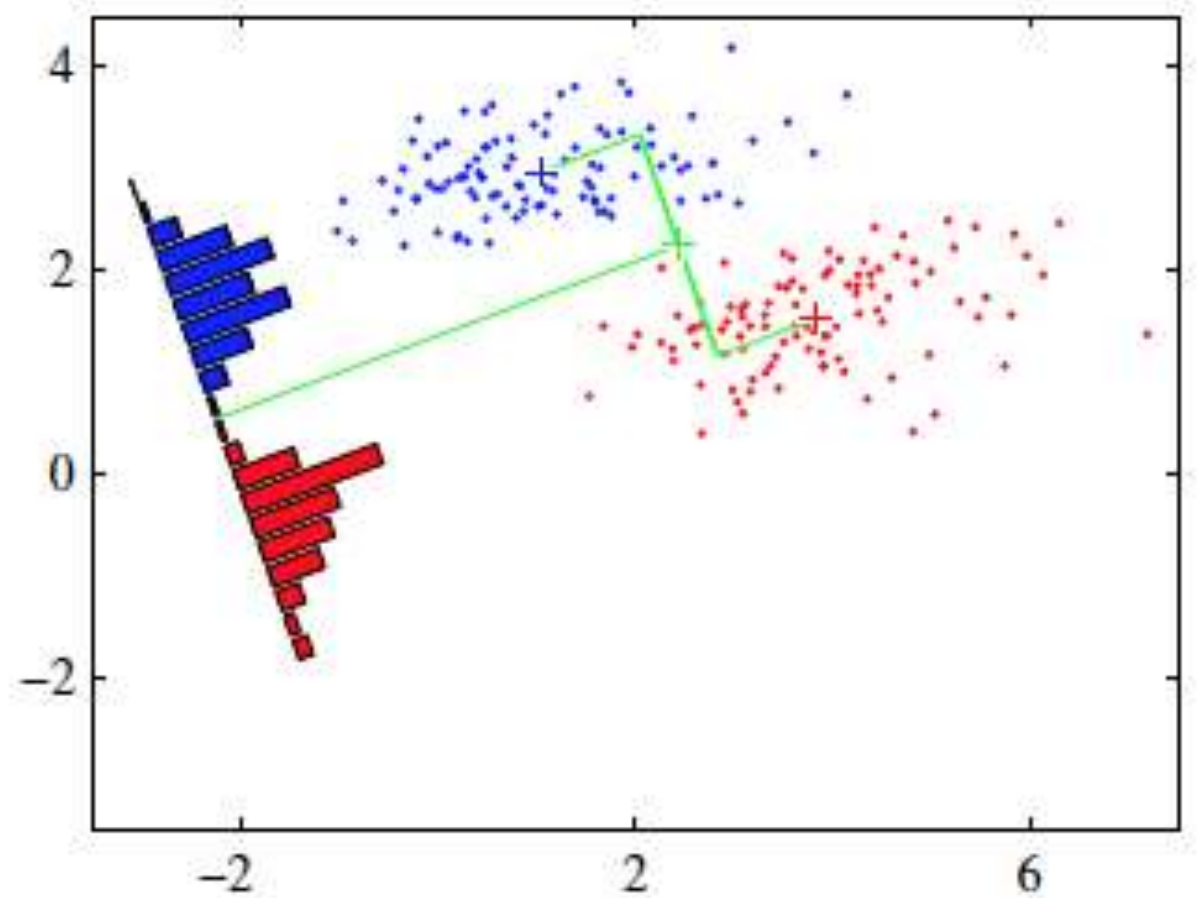
- ❖ Generalized Eigenvalue problem
- ❖ Eigenvectors of $\mathbf{S}_w^{-1} \mathbf{S}_b$

Linear Discriminant Analysis

Projecting on line joining means



Fisher Discriminant



PCA versus LDA

